Reading group notes: Kolmogorov-Arnold networks

Reading group of the Ashbury machine learning lab

1 Background

This work studies Kolmogorov-Arnold networks, a method that rethinks the collocation point. A practical system has to balance the cost of the symbolic formula against the quality of the basis function. Earlier Kolmogorov-Arnold network methods improved the residual activation but left the layer width unchanged. Recent work has made the basis function a central concern, because it decides how far Kolmogorov-Arnold network models can scale.

Combining Kolmogorov-Arnold networks with methods that target the spline grid is a promising direction. The grid interval also makes Kolmogorov-Arnold networks easier to inspect, since its behaviour can be read off per token. We expect the layer width to interact with the residual activation at larger scales, which we leave to future work. The results suggest that the grid interval is the main lever for quality at a fixed budget.

Despite steady progress, the interaction between the univariate function and the edge function is still poorly understood. Most prior methods treat the spline grid as fixed and tune the pruning threshold around it. This work studies Kolmogorov-Arnold networks, a method that rethinks the univariate function. Practitioners report that the symbolic formula is the first component to fail on long inputs.

2 Methods we read

The grid interval in Kolmogorov-Arnold networks is parameterised by a small linear layer with 64 hidden units. Gradients through the residual activation are clipped at a norm of 4.0. Kolmogorov-Arnold networks computes the basis function once per layer and reuses it for every head. Unlike a fixed-grid KAN, Kolmogorov-Arnold networks updates the symbolic formula only when the collocation point changes.

Gradients through the residual activation are clipped at a norm of 6.0. During inference Kolmogorov-Arnold networks caches the symbolic formula, so each new token costs a constant amount of work. Each block of Kolmogorov-Arnold networks applies the pruning threshold after a short convolution of width 2. The symbolic formula in Kolmogorov-Arnold networks is parameterised by a small linear layer with 512 hidden units.

A coefficient controls how strongly the grid interval depends on the residual activation; we set it to 0.8 throughout. The residual activation is initialised so that Kolmogorov-Arnold networks starts training close to an MLP. Unlike an MLP, Kolmogorov-Arnold networks updates the spline grid only when the residual activation changes. Kolmogorov-Arnold networks computes the layer width once per layer and reuses it for every head.

3 Results we discussed

Ablating the pruning threshold lowers the the Feynman equations score by 1.1 points, the largest drop among all components. Kolmogorov-Arnold networks scores 67.9% on a PDE suite, compared with 65.6% for an MLP. We train every model for 60B tokens with the same data order and report the mean of three seeds. On a tabular regression suite, Kolmogorov-Arnold networks improves over a Fourier feature network by 1.8 points.

Throughput is measured with a batch size of 512 on a single accelerator. With the symbolic formula frozen, Kolmogorov-Arnold networks still recovers most of its improvement on a tabular regression suite. Variance across seeds stays below 0.9 points for every configuration we report. Ablating the univariate function lowers the a tabular regression suite score by 5.5 points, the largest drop among all components.

With the collocation point frozen, Kolmogorov-Arnold networks still recovers most of its improvement on the Feynman equations. On a tabular regression suite, Kolmogorov-Arnold networks improves over an MLP by 4.5 points. Ablating the basis function lowers the a tabular regression suite score by 2.1 points, the largest drop among all components. Throughput is measured with a batch size of 16 on a single accelerator.

4 Open questions

Combining Kolmogorov-Arnold networks with methods that target the grid interval is a promising direction. A limitation of Kolmogorov-Arnold networks is that the collocation point must be tuned separately for each model size. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The results suggest that the basis function is the main lever for quality at a fixed budget.

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